

PRABHU SHANKAR

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SKILLS

Programming & Frameworks: Python (Pandas, NumPy, Scikit-learn, Boto3), SQL, Oracle PL/SQL, Java, Shell Script, Excel

Data Engineering & Orchestration: ETL/ELT Pipelines, dbt, Apache Airflow, Apache Spark (PySpark, Spark SQL), Kafka, Hadoop, Docker, Terraform, Data Pipeline Automation, Informatica

Cloud & Big Data Platforms: AWS (S3, EMR, Glue, Redshift, Lambda, Kinesis, RDS, IAM), Snowflake, Databricks, DynamoDB, Airbyte, Infrastructure as Code (IaC), CI/CD(Git, GitHub Actions), Azure Data Factory, KubeFlow

Data Architecture & Analytics: Data Modeling (Star Schema, Medallion, Snowflake), Data Warehousing, Distributed Data Processing, Optimization, Data Governance, Power BI, Tableau, Real-Time Streaming, Monitoring & Logging

EXPERIENCE

Data Engineering Academy

Brooklyn, NY

Data Engineer

Jul 2025 - Present

- Designed and optimized batch ETL pipelines using **AWS Glue, SQL, and Oracle PL/SQL**, reducing job runtime from 90 to 40 minutes and improving overall production data processing efficiency by 55%.
- Implemented near real-time ingestion using **Amazon Kinesis Firehose** to stream structured datasets into S3 and **Snowflake**, reducing reporting latency by 35% and enabling faster downstream analytics consumption.
- Developed **Spark (PySpark)** transformation pipelines on **Databricks**, applying partitioning, caching, and query optimization techniques to improve processing performance by 30% across high-volume production datasets.
- Built **dimensional data models** using star schema and **SCD Type 2** techniques in Snowflake, enabling reliable KPI reporting and improving dashboard accuracy across more than 20 business analytics use cases.
- Integrated **dbt** with **Snowflake** to standardize ELT transformations, implement automated testing, and enforce data quality checks, reducing downstream reporting discrepancies by 25% across analytics workflows.
- Led a **5-member agile data engineering team**, improving release velocity 15% and achieving 90% on-time deployments by enforcing ETL standards and optimizing AWS-based CI/CD pipeline workflows.

Mercer University

Atlanta, GA

Research Assistant

Aug 2024 - May 2025

- Engineered ELT pipelines on **Amazon Redshift** for \$2B AUM datasets, implementing caching strategies, distribution and sort keys, and Python-based anomaly detection to reduce daily ETL latency by 45 minutes.
- Built **Kafka-based ingestion pipelines** with consumer groups and offset management to stream high-volume datasets into warehouse staging tables, improving data freshness and reducing reporting latency by 20%.
- Optimized warehouse query performance by 11% through **SARGable SQL tuning**, partition pruning, and indexing strategies, improving reporting efficiency and reducing compute costs across analytics workloads.
- Designed normalized and dimensional data models using **star schema techniques**, enabling scalable data warehousing and accelerating KPI forecasting dashboards by 30% through optimized transformation workflows.
- Reduced storage footprint by 200TB by introducing **feature-engineered columns** and refining retention policies within ML pipelines, improving processing efficiency and generating \$15K in biannual revenue impact.

Goldman Sachs - Promotions

Bengaluru, India

Senior Data Analyst

Jan 2022 - Dec 2023

- Reduced **data model loading time by 25+ hours** weekly by implementing 3NF-compliant warehouse schemas in Databricks and optimizing ELT pipelines using **PySpark, SQL, and Terraform**-driven infrastructure automation.
- Enhanced ingestion pipeline efficiency by 10% by implementing **CI/CD automation** with **Git** and **Spark**-based transformations loading into **Amazon Redshift**, enabling bi-weekly zero-downtime production deployments.
- Led an 8-member agile team through full **SDLC lifecycle**, defining technical architecture on AWS, developing **SQL**-based transformations, and accelerating sprint deployments by 25% with 98% on-time delivery.

Analyst

May 2020 - Dec 2021

- Saved \$250K annually by implementing scalable batch processing architecture using **Snowflake data warehousing, Airbyte** ingestion, and optimized **PySpark** transformations while maintaining strict SLA compliance for high-volume analytics workloads.
- Improved pipeline throughput by 20% by redesigning cloud-based data architecture using **Inmon data modeling**, partitioning strategies, and **SQL performance tuning**, enhancing freshness across 15+ executive Tableau dashboards.
- Reduced manual intervention by 60% by developing **Airflow-orchestrated ETL** pipelines with dbt transformations and AI-driven log monitoring, implementing automated **data quality checks** and structured error handling frameworks.
- Improved ETL execution efficiency by 20% by modernizing **AWS**-based data infrastructure and optimizing **Databricks** transformations, enabling faster data availability and reliable reporting for 15+ production dashboards.

CERTIFICATIONS

- dbt
- Alteryx Designer
- Databricks Fundamentals
- Lean Six Sigma

EDUCATION

Mercer University, Atlanta, GA - *Master of Science in Data Analytics*

May 2025

- *Relevant courses: Data systems, Business Analytics, Visualization, Statistics, and Workforce Insights*
- *Represented the college in the [Big Data Health Competition](#), ranking in the top 5%*

PES University, Bengaluru, IN - *Bachelor of Technology in Electronics and Communication*

Aug 2020

- *Built a gait-based biometric [recognition model](#) with SSFE algorithm and neural network, achieving 80% accuracy*

PROJECTS AND CONTRIBUTIONS

[Wistia Media Analytics Pipeline](#) | Python, AWS S3, Apache Airflow, Pandas, Parquet

- Built a production-grade medallion ETL pipeline (Bronze -> Silver -> Gold) ingesting Wistia API data daily, achieving 68% storage reduction via JSON-to-Parquet compression, processing ~85,000 records across 350 media objects per run with <1.2-minute end-to-end runtime and <1% failure rate through Airflow-orchestrated retry logic

[Data Pipeline Development](#) | Python, Apache, dbt, Azure

- Developed a containerized ETL pipeline to extract, clean, and store football stadium data into ADL, processing 100+ stadium records per run and reducing manual data collection time by 90%, enabling win-percentage analytics that helped the college team secure 10 to 12 additional match wins per season

[Social Media API Development](#) | FastAPI, PostgreSQL, SQLAlchemy, JWT

- Engineered a RESTful API with secure authentication, post management, and voting system, handling 1,000+ concurrent requests with <200ms latency and improving developer onboarding by 50% through automated docs and modularized routing

[Presto DB - Open Source Contributor](#) | Java, SQL, Presto Web UI

- Identified and fixed a SQL preprocessing bug in the Presto Web UI affecting 9,000–10,000 weekly users; fix was reviewed, approved, and merged into the main branch of the 16.7K-star repository